Building and evaluation of a PBPK model for Atomoxetine in CYP2D6 phenotype and activity-score groups

Version	evaluation-OSP12.2
based on *Model Snapshot* and *Evaluation Plan*	https://github.com/Open-Systems-Pharmacology/Atomoxetine-Model/releases/tag/vevaluation
OSP Version	12.2
Qualification Framework Version	3.5

This evaluation report and the corresponding PK-Sim project file are filed at:

https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/

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1 Introduction

Atomoxetine is a selective norepinephrine reuptake inhibitor used for the treatment of attention-deficit/hyperactivity disorder. It is administered orally and is predominantly cleared by CYP2D6-mediated metabolism, which leads to clinically relevant exposure differences between CYP2D6 phenotype and activity-score groups.

This atomoxetine model is intended to describe plasma concentration-time profiles of atomoxetine across CYP2D6 phenotype and activity-score groups. It supports evaluation of CYP2D6 drug-gene interactions and interacting-drug scenarios where atomoxetine exposure is clinically relevant.

The whole-body PBPK model of atomoxetine was developed by Rüdesheim 2022 together with paroxetine and risperidone models to describe CYP2D6 activity score-dependent metabolism. The clinical data include oral single-dose and repeated-dose administration of atomoxetine as solution, capsule, or tablet, CYP2D6 phenotype or activity-score stratified groups, and studies with CYP2D6 inhibition or genetic activity differences.

The presented model includes the following features:

- atomoxetine as the evaluated parent compound,
- CYP2D6-mediated metabolism with activity-score-dependent k_{cat} values,
- CYP2C19-mediated metabolism as a pragmatic non-CYP2D6 CYP pathway,
- passive renal filtration,
- oral solution, capsule, and tablet administration in the evaluated clinical studies.

2 Methods

2.1 Modeling strategy

The general concept of building a PBPK model has previously been described by Kuepfer et al. (Kuepfer 2016). Relevant information on anthropometric and physiological parameters in adults was gathered from the literature and incorporated into PK-Sim as default values for adult simulations (Willmann 2007).

The applied activity and variability of plasma proteins and active processes integrated into PK-Sim are described in the publicly available PK-Sim Ontogeny Database or otherwise referenced for the specific process.

The atomoxetine model was developed as a single-parent compound model for oral atomoxetine pharmacokinetics. The model by Rüdesheim 2022 described CYP2D6 activity score-dependent metabolism by representing CYP2D6 metabolic capacity with activity-score-dependent k_{cat} values. CYP2D6-independent clearance was represented through a CYP2C19 pathway and passive renal filtration.

Clinical studies used for model building covered oral atomoxetine administration and included solution, capsule, and tablet dosing where available. Model-building studies informed oral absorption, CYP2D6-independent clearance in poor metabolizers, and CYP2D6-dependent clearance in extensive metabolizers or activity-score groups. Model verification used independent oral study arms, including CYP2D6 poor-, normal-, intermediate-, and higher-activity groups.

The model-building data covered the key structural questions of the atomoxetine model. Oral solution and capsule data supported absorption parameterization, poor-metabolizer data supported the CYP2D6-independent clearance pathway, and extensive-metabolizer or activity-score data supported CYP2D6-dependent clearance. Verification data were then used to evaluate whether the final model described independent concentration-time profiles without additional study-specific fitting.

The represented processes are CYP2D6-mediated metabolism, CYP2C19-mediated metabolism, plasma protein binding, and passive renal filtration. Atomoxetine metabolites were not explicitly implemented because only limited plasma concentration-time profiles for 4-hydroxyatomoxetine were available. CYP2C19 was used as a surrogate pathway for smaller CYP contributions reported *in vitro* and should be interpreted as a pragmatic clearance pathway rather than as a claim that CYP2C19 is the only non-CYP2D6 metabolic route.

For the CYP2D6 drug-gene interaction model, CYP2D6-dependent clearance was represented with Michaelis-Menten kinetics. CYP2D6 k_{cat} values were optimized for extensive-metabolizer or activity-score groups, set to zero for poor metabolizers, and scaled for other activity scores using the activity-score framework from Rüdesheim 2022.

The evaluated applications are oral single-dose and repeated-dose applications using solution, capsule, or tablet dosing. The major proteins and processes represented explicitly are CYP2D6, CYP2C19, plasma protein binding, and passive renal filtration. CYP2D6 is the primary mechanistic driver for phenotype and activity-score comparisons, while CYP2C19 and renal filtration account for non-CYP2D6 elimination.

Model performance was assessed with goodness-of-fit diagnostics and concentration-time profiles for the model-building and verification sets. The evaluation emphasizes the separation between CYP2D6 phenotype or activity-score groups because this is the central intended use of the atomoxetine model.

Details about input data are provided in Section 2.2. Details about the structural model and assumptions are provided in Section 2.3.

2.2 Data used

2.2.1 In vitro and physicochemical data

The table below summarizes the drug-dependent inputs documented for the atomoxetine model.

Parameter	Unit	Value	Source	Description
Atomoxetine MW	g/mol	255.35	Zhong 2013	Compound size used in concentration conversions.
Atomoxetine pK_a base	-	9.80	Swain 2012	Basic ionization constant.
Atomoxetine solubility, pH 7.4	mg/mL	10.29	Swain 2012	Solubility input.
Atomoxetine f_u	%	1.30	Yu 2016	Plasma binding input.
Atomoxetine CYP2D6 K_m	µmol/L	2.30	Ring 2002	CYP2D6 affinity held constant across activity scores.
Atomoxetine CYP2D6 k_{cat} EM	1/min	37.44	Optimized; Ring 2002	activity-score-dependent CYP2D6 clearance.
Atomoxetine CYP2D6 k_{cat} PM	1/min	0	Assumed	CYP2D6 poor-metabolizer activity set to zero.
Atomoxetine CYP2C19 K_m	µmol/L	83.00	Ring 2002	CYP2C19 affinity parameter.
Atomoxetine CYP2C19 k_{cat}	1/min	165.23	Optimized; Ring 2002	Non-CYP2D6 metabolic clearance surrogate.
Atomoxetine logP	-	3.49	Optimized	Distribution input.
GFR fraction	-	1.00	Assumed	Passive glomerular filtration fraction.
Partition coefficients	-	Diverse	Calculated	Partition coefficient method.
Cellular permeabilities	cm/min	0.32	Calculated	Cellular permeability method.
Specific intestinal permeability	cm/min	5.23E-05	Optimized	Oral absorption parameter.

2.2.2 Clinical data

The evaluation uses 14 observed-data profiles for atomoxetine in peripheral venous blood plasma. The evaluation plan assigns 6 simulations to model building and 8 simulations to model verification.

Model-building clinical data:

Publication	Arm / Treatment / Information used for model building
Nakano 2016	Plasma PK profiles in adults after oral, single-dose administration of 50 mg atomoxetine as solution or capsule with CYP2D6 extensive-metabolizer status.
Byeon 2015	Plasma PK profile in adults after oral, single-dose administration of 40 mg atomoxetine with CYP2D6 AS = 2 status.
Kim 2018	Plasma PK profile in adults after oral, single-dose administration of 20 mg atomoxetine with CYP2D6 AS = 2 status.
Sauer 2003	Plasma PK profiles in adults after repeated oral administration of 20 mg atomoxetine with CYP2D6 extensive- and poor-metabolizer status.

Model-verification clinical data:

Publication	Arm / Treatment / Information used for model verification
Belle 2002	Plasma PK profiles in adults after oral, once daily administration of 20 mg atomoxetine with CYP2D6 extensive-metabolizer status.
Cui 2007	Plasma PK profiles in adults after oral administration of 40 mg and 80 mg atomoxetine with CYP2D6 AS = 1 status.
Byeon 2015	Plasma PK profiles in adults after oral, single-dose administration of 40 mg atomoxetine with CYP2D6 AS = 0.5 and AS = 1.25 status.
Kim 2018	Plasma PK profiles in adults after oral, single-dose administration of 20 mg atomoxetine with CYP2D6 AS = 0.5 status.
Todor 2016	Plasma PK profiles in adults after oral, single-dose administration of 25 mg atomoxetine with CYP2D6 normal- and poor-metabolizer status.

2.3 Model parameters and assumptions

2.3.1 Absorption

The model includes oral solution, capsule, and tablet applications. Oral absorption was described using PK-Sim permeability and formulation settings. The `Specific intestinal permeability` was optimized to 5.23E-05 cm/min based on oral concentration-time data.

The oral solution study by Nakano 2016 provided information-rich data for absorption, while capsule and tablet study arms were used to evaluate oral absorption under clinically used formulations.

No separate dissolution model was introduced for the immediate-release applications. Differences between solution and solid oral applications were represented through the application and formulation settings used for each simulation. This keeps the absorption model compact while preserving the clinically relevant distinction between solution, capsule, and tablet dosing.

Absorption parameterization was evaluated against both model-building and verification study arms. This is relevant because the same intestinal permeability value is used across phenotype and activity-score groups, so absorption should not compensate for differences that are mechanistically attributed to CYP2D6 activity.

2.3.2 Distribution

Atomoxetine is highly bound to plasma proteins. A fraction unbound of 1.30% was used as summarized in Section 2.2.1.

Lipophilicity was optimized to logP 3.49. Partition coefficients were calculated with the Berezhkovskiy method, and cellular permeability was calculated by PK-Sim.

The optimized lipophilicity was used together with plasma protein binding to describe tissue distribution and systemic exposure. Because atomoxetine is highly protein bound, changes in fraction unbound would directly affect the unbound concentration available for hepatic metabolism. The value was therefore kept consistent with the documented physicochemical and binding data rather than fitted separately by CYP2D6 group.

Distribution was not stratified by CYP2D6 phenotype. Differences between poor, normal, and higher-activity groups are represented through metabolic capacity, while the same compound distribution assumptions are applied to all adult simulations.

2.3.3 Metabolism and Elimination

Two metabolic pathways and passive renal filtration are represented in the model.

- CYP2D6

CYP2D6 is the dominant pathway for atomoxetine clearance in normal metabolizers. CYP2D6 metabolism is represented by Michaelis-Menten kinetics. CYP2D6 poor-metabolizer k_{cat} is set to zero, and non-zero activity-score groups use optimized or activity-score scaled k_{cat} values.

The same CYP2D6 K_m is used across CYP2D6 groups. Activity-score effects are represented by changes in k_{cat}, which is consistent with the activity-score framework used for the CYP2D6 model evaluation. This separates enzyme affinity from phenotype-dependent or activity-score-dependent metabolic capacity.

- CYP2C19

CYP2C19 was implemented as a surrogate pathway for minor non-CYP2D6 CYP contributions reported *in vitro*. CYP2C19 K_m was taken from Ring 2002, and CYP2C19 k_{cat} was optimized.

This pathway contributes to residual oxidative metabolism in poor metabolizers and prevents the CYP2D6-independent part of clearance from being represented only by renal elimination. The CYP2C19 pathway should therefore be interpreted as a pragmatic model pathway for minor CYP-mediated clearance, not as a claim that CYP2C19 is the only non-CYP2D6 metabolic route.

- Renal elimination

Passive renal filtration is represented with a `GFR fraction` of 1.

Renal elimination is a minor structural pathway relative to CYP-mediated metabolism. It was retained as passive filtration using the model fraction unbound and default adult renal physiology.

3 Results and Discussion

The PBPK model for atomoxetine was developed and evaluated with clinical pharmacokinetic data after oral administration. The evaluation covers single-dose and repeated-dose oral administration, solution, capsule, and tablet dosing, and CYP2D6 extensive-, normal-, intermediate-, and poor-metabolizer settings represented by genotype, phenotype, or activity score.

The model-building data supported the oral absorption description, CYP2D6-mediated clearance, CYP2D6-independent metabolism, passive renal filtration, and activity-score-dependent clearance implementation. Model-building studies included solution and capsule data from Nakano 2016, CYP2D6 poor- and extensive-metabolizer data from Sauer 2003, and activity-score stratified data from Byeon 2015 and Kim 2018. Verification included independent oral studies from Belle 2002, Cui 2007, Byeon 2015, Kim 2018, and Todor 2016.

The model quantifies CYP2D6-mediated metabolism, CYP2C19-mediated non-CYP2D6 metabolism, and passive glomerular filtration. The interpretation of model performance requires separate consideration of CYP2D6 poor-metabolizer profiles and profiles with residual or higher CYP2D6 activity. Poor-metabolizer profiles mainly test the non-CYP2D6 clearance pathway and renal filtration. Normal- and higher-activity groups additionally test whether increased CYP2D6 k_{cat} values describe the observed activity-score-dependent exposure decrease.

The next sections show:

1. the final model input parameters for the building blocks: Section 3.1.
2. the overall goodness of fit: Section 3.2.
3. simulated vs. observed concentration-time profiles for the clinical studies used for model building and for model verification: Section 3.3.

The merged GOF diagnostic over the included concentration observations gives a GMFE of 1.28 for atomoxetine. This value shows good population-level agreement across the represented studies, but it should still be interpreted together with the concentration-time profiles because the dataset combines different CYP2D6 activity groups, formulations, and repeated-dose behavior.

Rüdesheim 2022 reported good atomoxetine base-model performance. Across atomoxetine base-model evaluations, the reported mean GMFE was 1.20 for AUClast and 1.18 for Cmax. For plasma concentration values, 88.9% of atomoxetine predictions were within two-fold of the corresponding observed values. These published metrics are consistent with the merged concentration GMFE in Section 3.2.

The DGI ratio evaluation is important for the intended use of this model. The included simulations cover studies by Cui 2007, Byeon 2015, Kim 2018, Sauer 2003, and Todor 2016. Rüdesheim 2022 reported that 5 of 5 AUClast ratios and 5 of 5 Cmax ratios were within the prediction success limits, with GMFE values of 1.25 for AUClast ratios and 1.28 for Cmax ratios.

The concentration-time profiles should be reviewed by CYP2D6 activity group. Since absorption and distribution parameters are shared across activity groups, systematic differences by group would indicate clearance-model limitations rather than formulation-specific adjustment. The current diagnostics support the use of the model for adult oral atomoxetine simulations across the represented CYP2D6 phenotype and activity-score range.

The model is adequate for adult oral atomoxetine simulations within the represented doses, formulations, and CYP2D6 groups. Remaining interpretation should be cautious for unrepresented formulations, pediatric populations, and scenarios where non-CYP2D6 pathways are expected to differ from the adult populations used for model evaluation.

3.1 Atomoxetine final input parameters

The final Atomoxetine input parameter table summarizes the drug-dependent model parameters used for the evaluated simulations.

Compound: Atomoxetine

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	10.29 mg/ml	Publication-Literature value in, Swain 2012 [47]	Measurement	True
Reference pH	7.4	Publication	Measurement	True
Lipophilicity	3.4885087539 Log Units	Parameter Identification-Parameter Identification-Optimized	LogP	True
Lipophilicity	3.65 Log Units		Mohan 2015	False
Lipophilicity	3.81 Log Units	Publication-Literature value in, Swain 2012 [47]	ChemAxon	False
Fraction unbound (plasma, reference value)	1.3 %	Publication-In Vivo-Literature value in, Yu 2016 [56]	Measurement	True
Fraction unbound (plasma, reference value)	3.9 %	Unknown-Watanabe 2018	Calculated	False
Specific intestinal permeability (transcellular)	5.23E-05 cm/min	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Is small molecule	Yes			
Molecular weight	255.35 g/mol	Publication-Literature value in, Zhong et al. 2013 [57]		
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Berezhkovskiy
Cellular permeabilities	PK-Sim Standard

Processes

Metabolizing Enzyme: CYP2D6-Ring 2002 (4-hydroxylation)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	115 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	2.3 µmol/l	Publication-In Vitro-Literature value in, Ring et al. 2002 [39]
kcat	37.4392066898 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=0

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	2.3 µmol/l	Publication-In Vitro-Same CYP2D6 Km as base model, Ring et al. 2002 [39]

Systemic Process: Glomerular Filtration-Assumed

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	Other-Assumed

Metabolizing Enzyme: CYP2D6-AS=2

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	2.3 µmol/l	Publication-In Vitro-Same CYP2D6 Km as base model, Ring et al. 2002 [39]
kcat	102.25 1/min	Parameter Identification-Parameter Identification-Optimized AS=2 baseline, Rüdesheim et al. 2022, main article Table 2

Metabolizing Enzyme: CYP2D6-AS=0.5

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	2.3 µmol/l	Publication-In Vitro-Same CYP2D6 Km as base model, Ring et al. 2002 [39]
kcat	16.87 1/min	Other-Manual Fit-CYP2D6 activity-score scaling, Rüdesheim et al. 2022, main article Table 2

Metabolizing Enzyme: CYP2D6-AS=1.25

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	2.3 µmol/l	Publication-In Vitro-Same CYP2D6 Km as base model, Ring et al. 2002 [39]
kcat	53.68 1/min	Other-Manual Fit-CYP2D6 activity-score scaling, Rüdesheim et al. 2022, main article Table 2

Metabolizing Enzyme: CYP2C19-Ring 2002

Molecule: CYP2C19

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	97 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	19 pmol/mg mic. protein	Unknown
Km	83 µmol/l	Publication-In Vitro-Literature value in, Ring et al. 2002 [39]
kcat	165.2291488831 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=1

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	2.3 µmol/l	Publication-In Vitro-Same CYP2D6 Km as base model, Ring et al. 2002 [39]
kcat	39.7 1/min	Other-Manual Fit-CYP2D6 activity-score scaling, Rüdesheim et al. 2022, main article Table 2

3.2 Atomoxetine diagnostic plots

The goodness-of-fit diagnostics combine all modeled compounds in one set of plots. Colors and symbols identify compounds consistently with the concentration-time profiles. Administration route, formulation, and model-building or verification status are not used to split the diagnostics.

Table 3-1: GMFE for Observed versus simulated concentration-time data for all modeled compounds

Group	GMFE
Atomoxetine	1.28

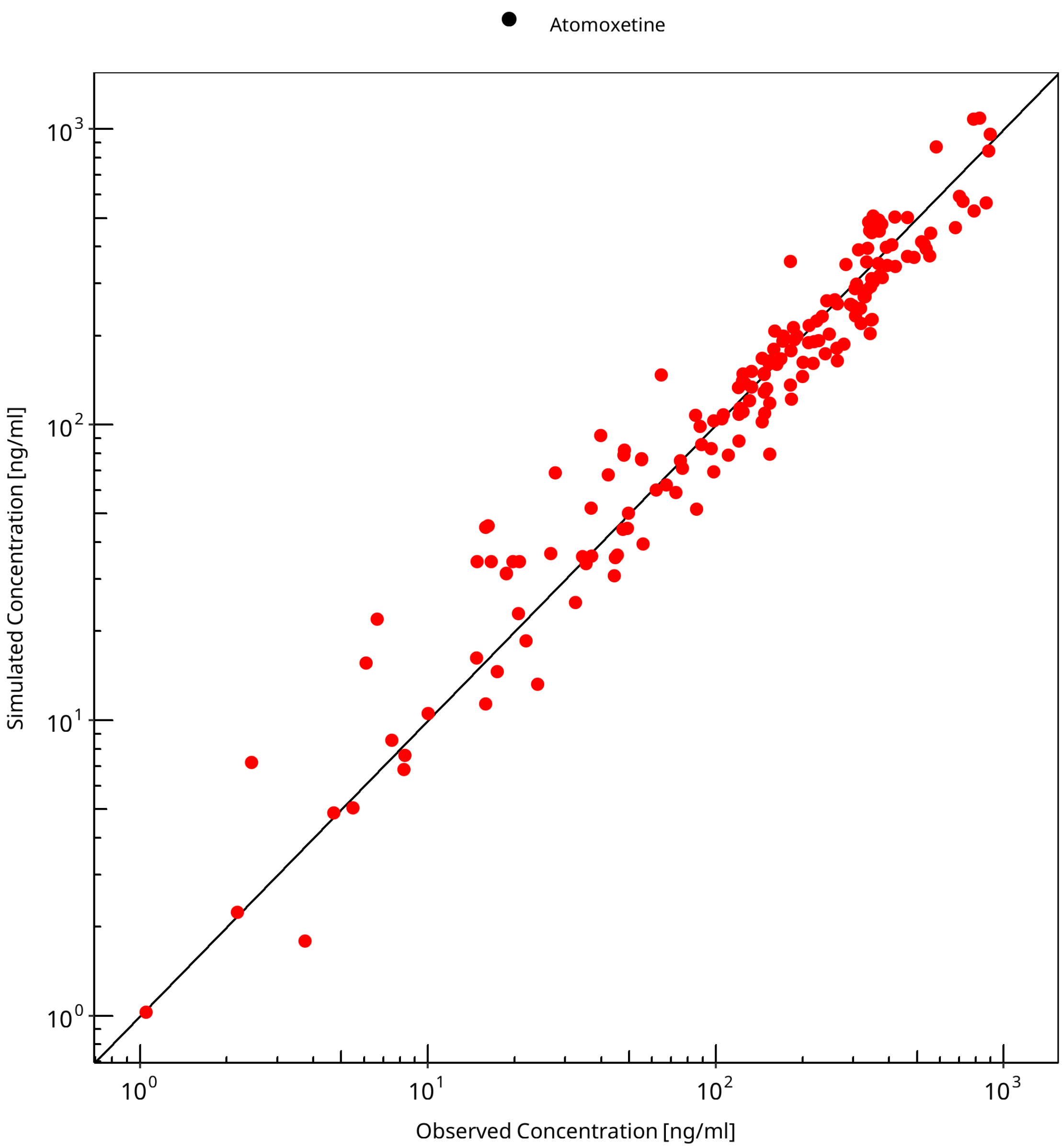

Figure 3-1: Observed versus simulated concentration-time data for all modeled compounds

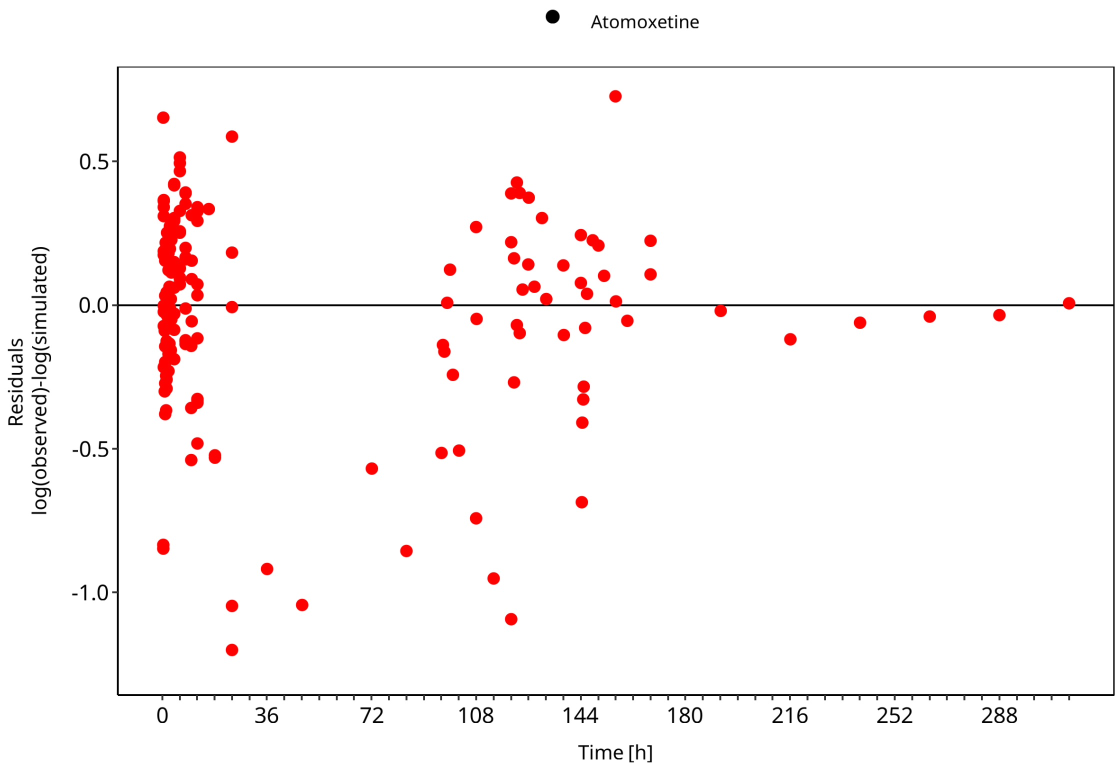

Figure 3-2: Observed versus simulated concentration-time data for all modeled compounds

3.3 Concentration-time profiles

Simulated versus observed concentration-time profiles of all data listed in Section 2.2.2 (Clinical data) are presented below.

3.3.1 Model Building

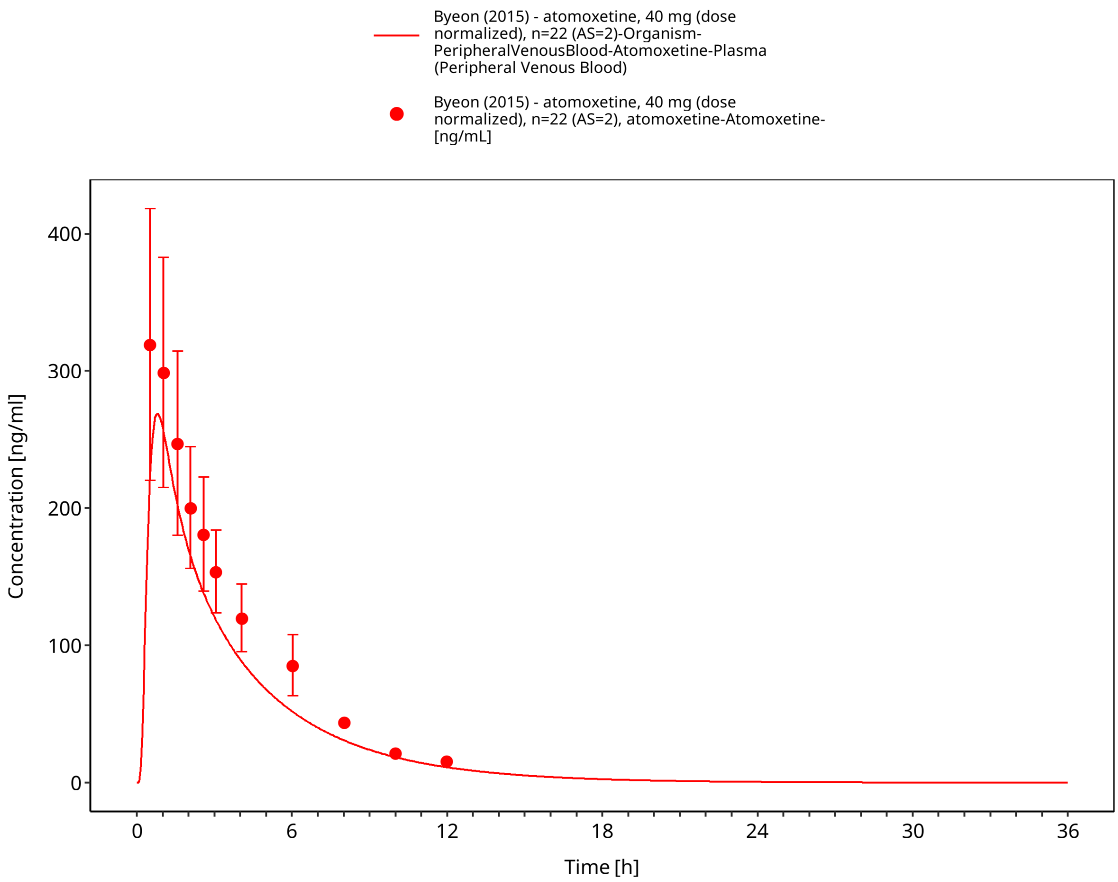

Figure 3-3: Time Profile Analysis

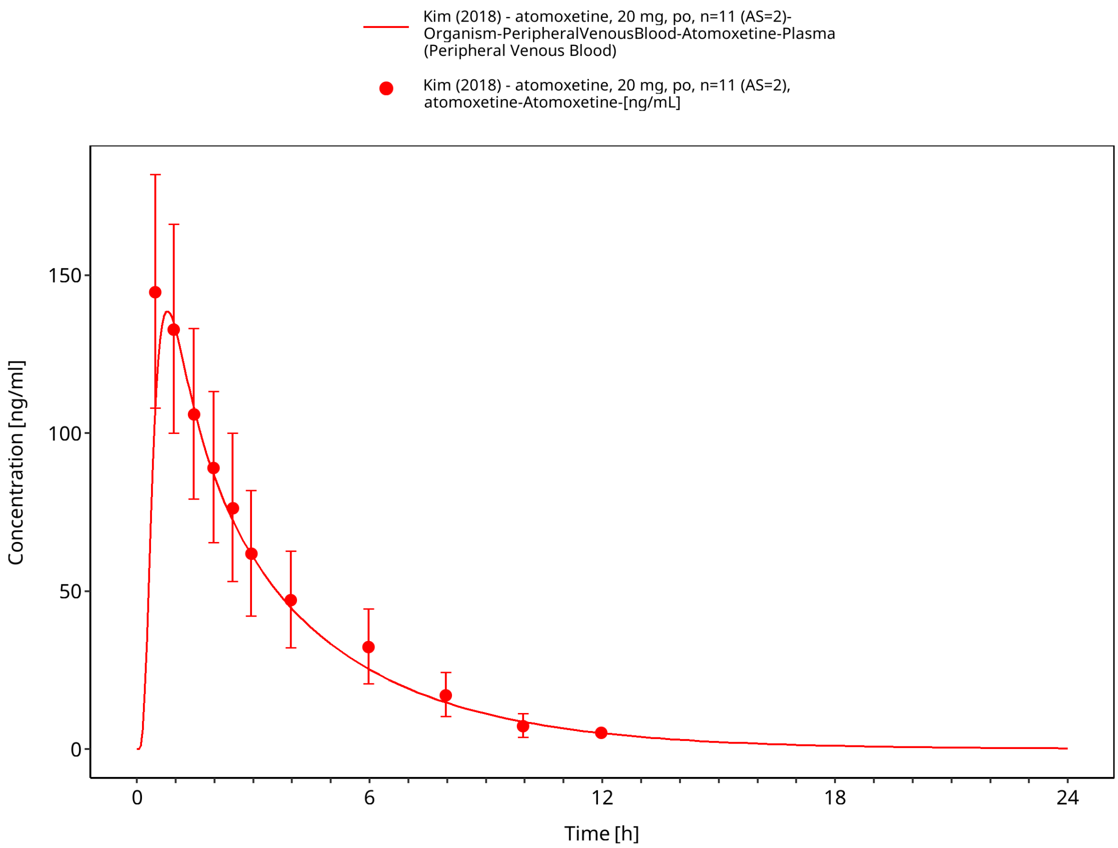

Figure 3-4: Time Profile Analysis

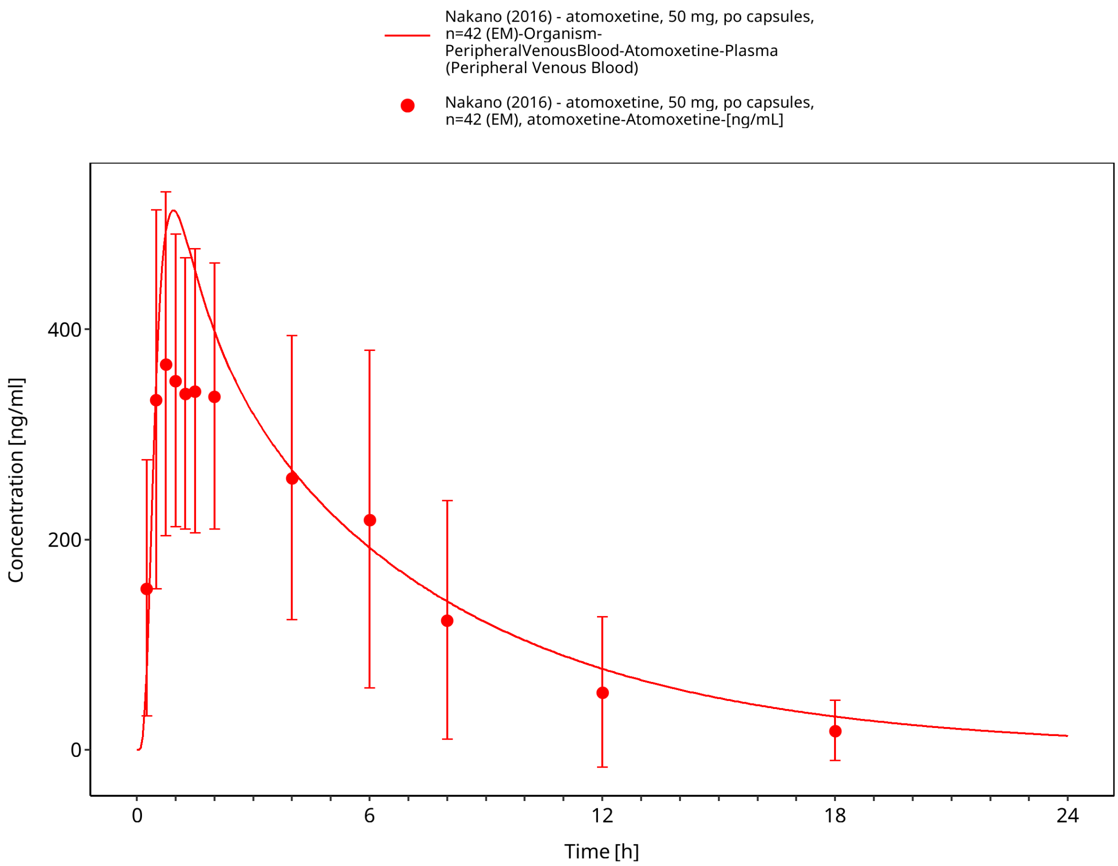

Figure 3-5: Time Profile Analysis

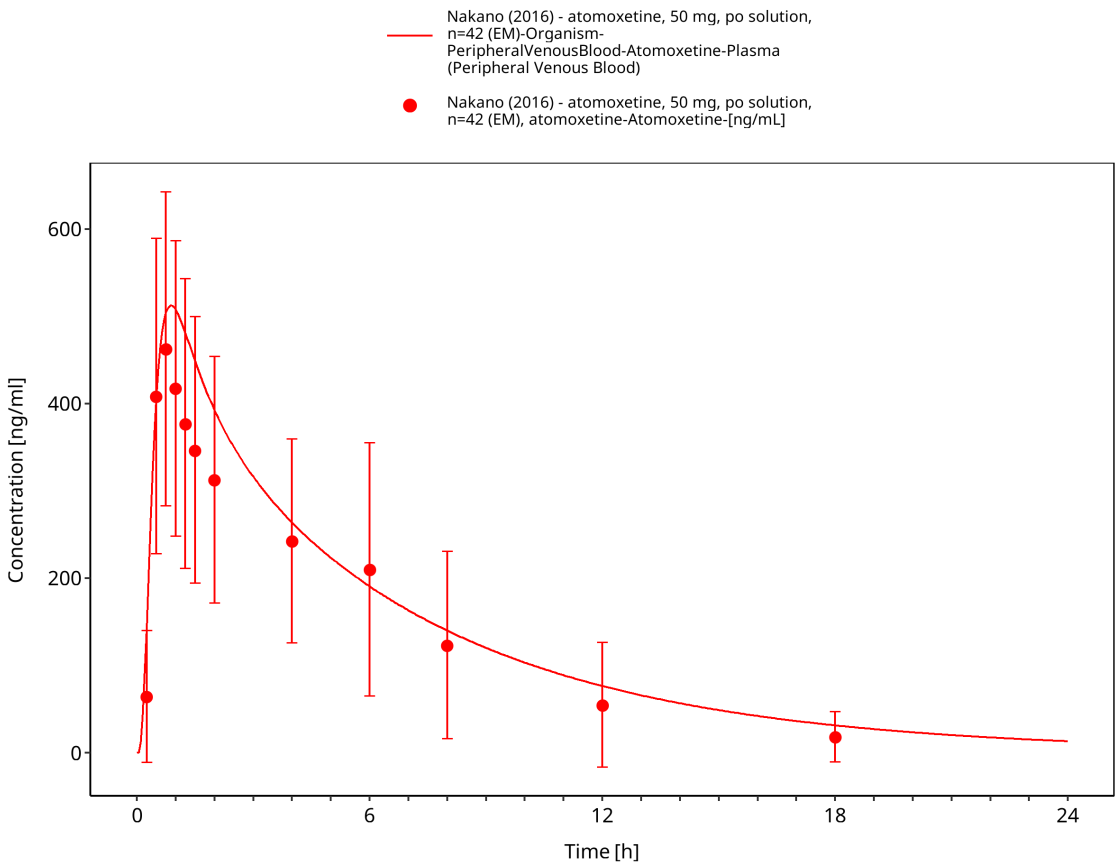

Figure 3-6: Time Profile Analysis

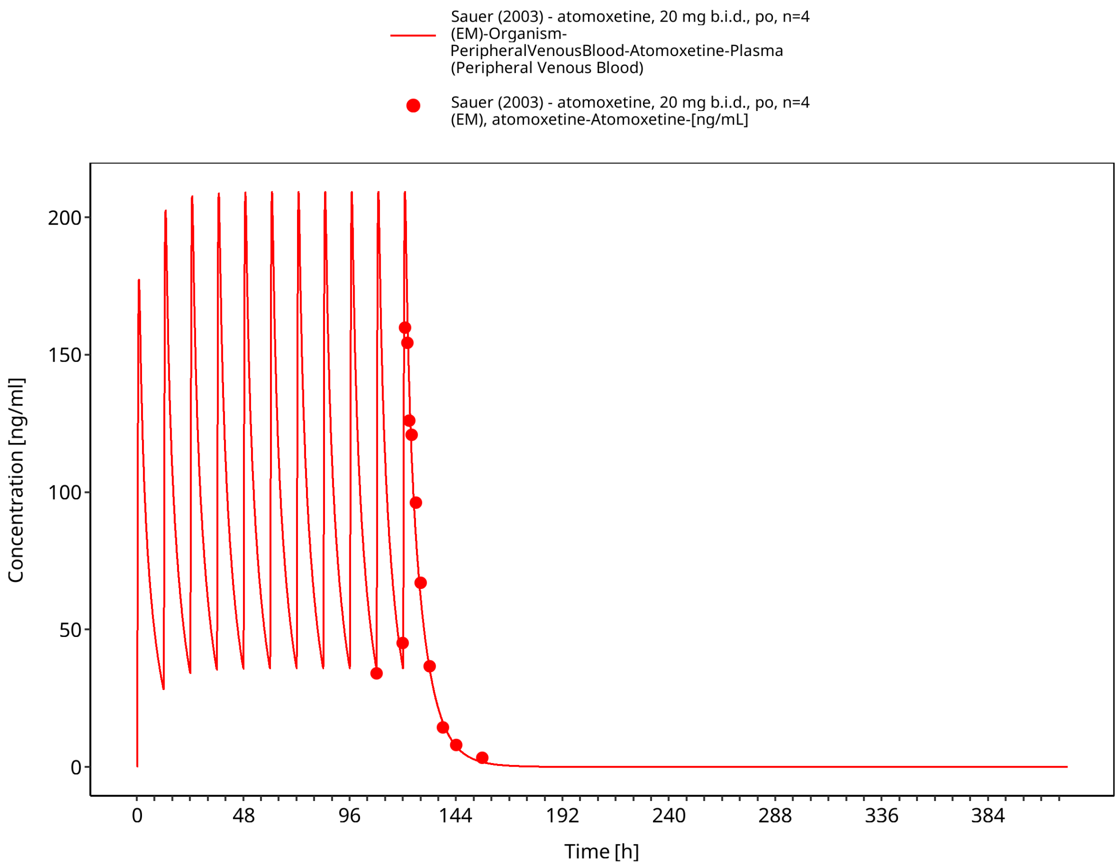

Figure 3-7: Time Profile Analysis

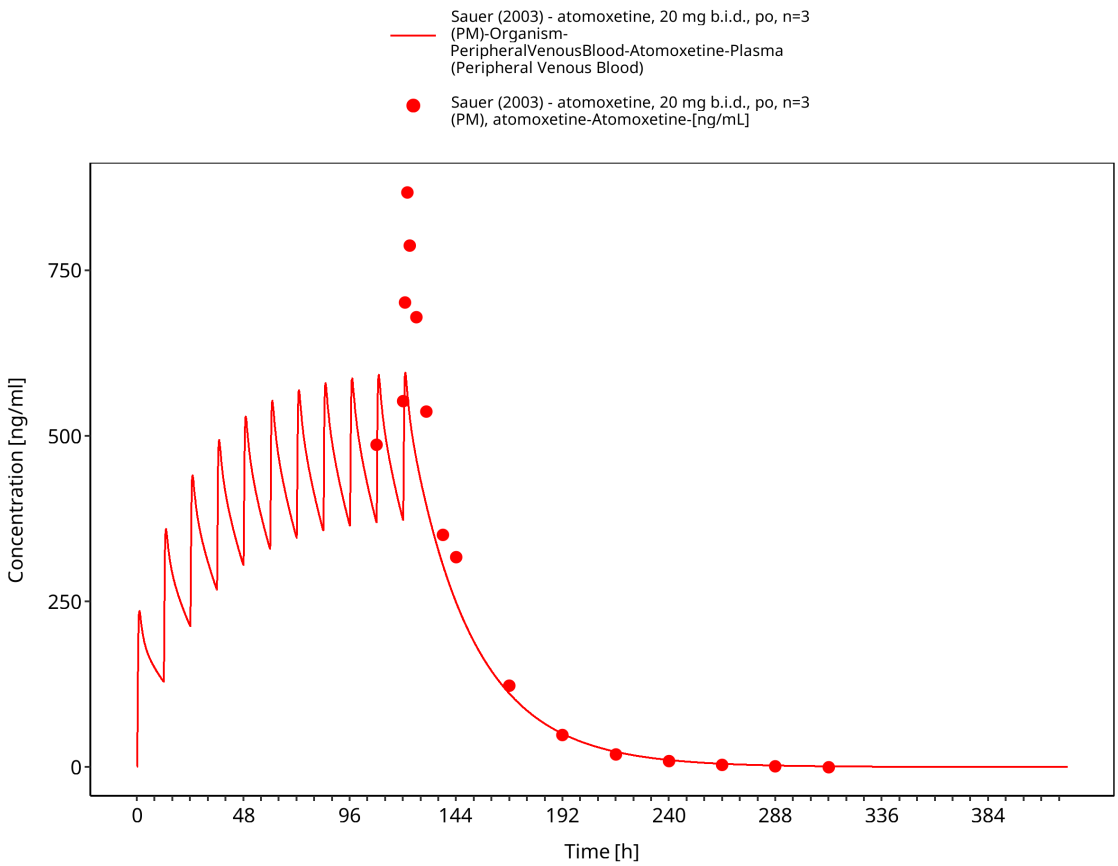

Figure 3-8: Time Profile Analysis

3.3.2 Model Verification

The following verification profiles compare the final model with independent atomoxetine studies across dose, formulation, and CYP2D6 activity groups. They should be interpreted together with the study-specific observed-data records and the merged goodness-of-fit diagnostic.

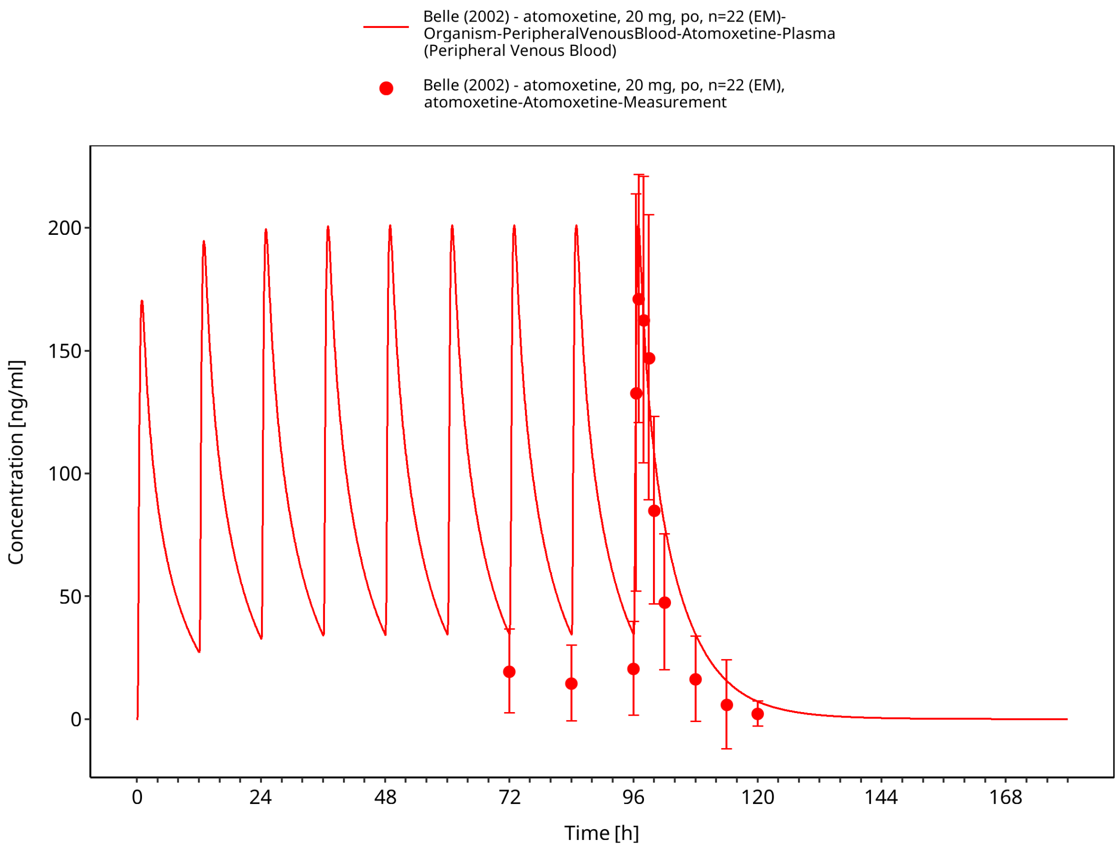

Figure 3-9: Time Profile Analysis

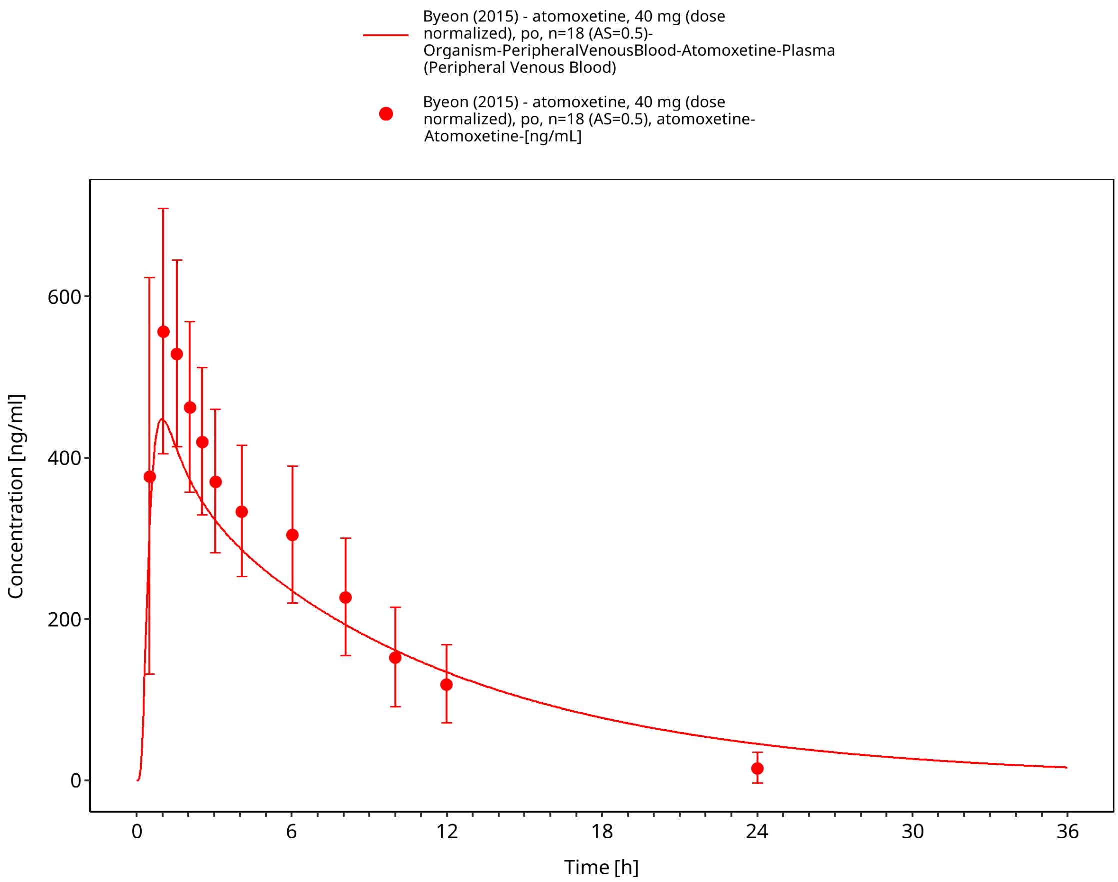

Figure 3-10: Time Profile Analysis

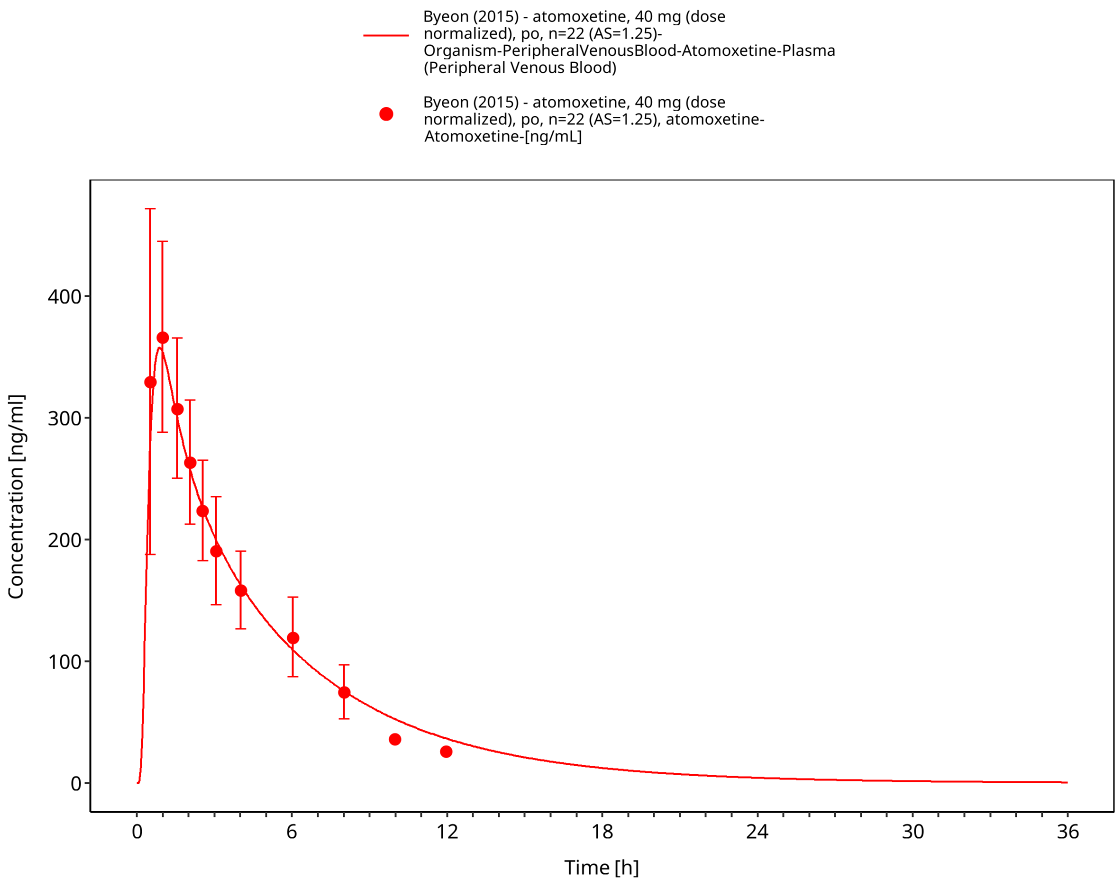

Figure 3-11: Time Profile Analysis

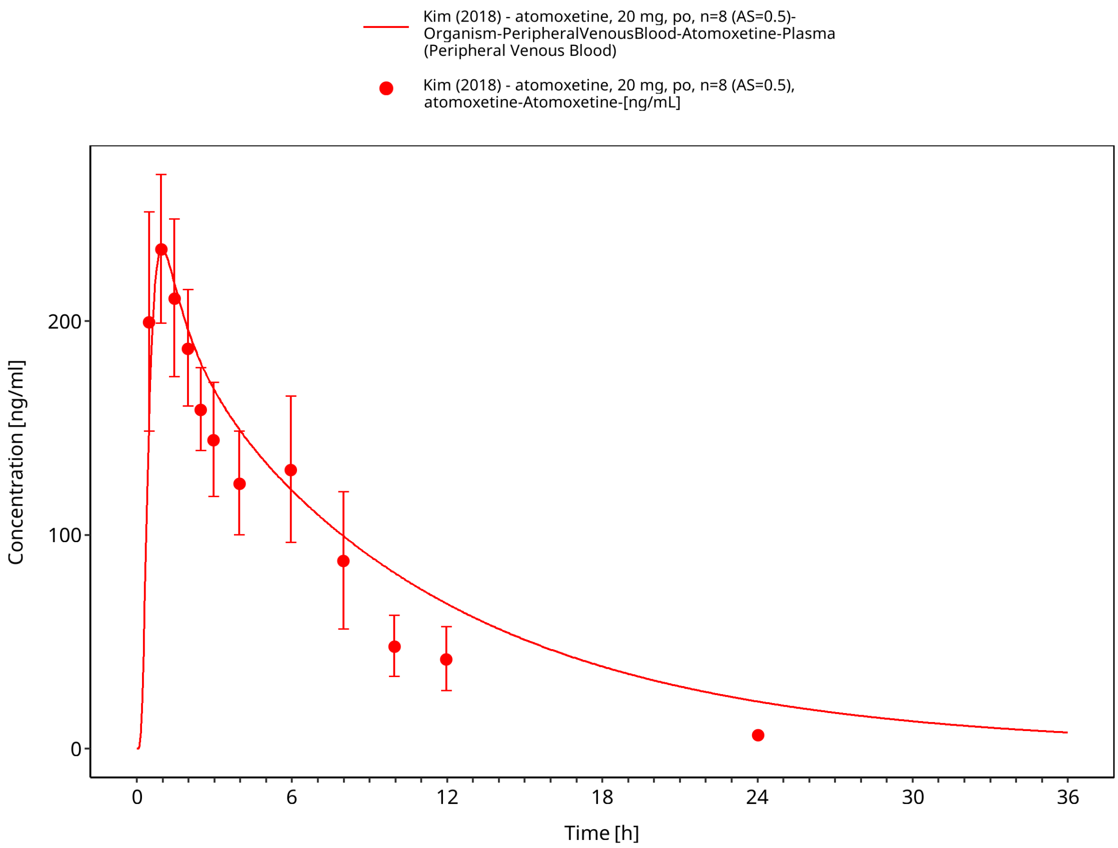

Figure 3-12: Time Profile Analysis

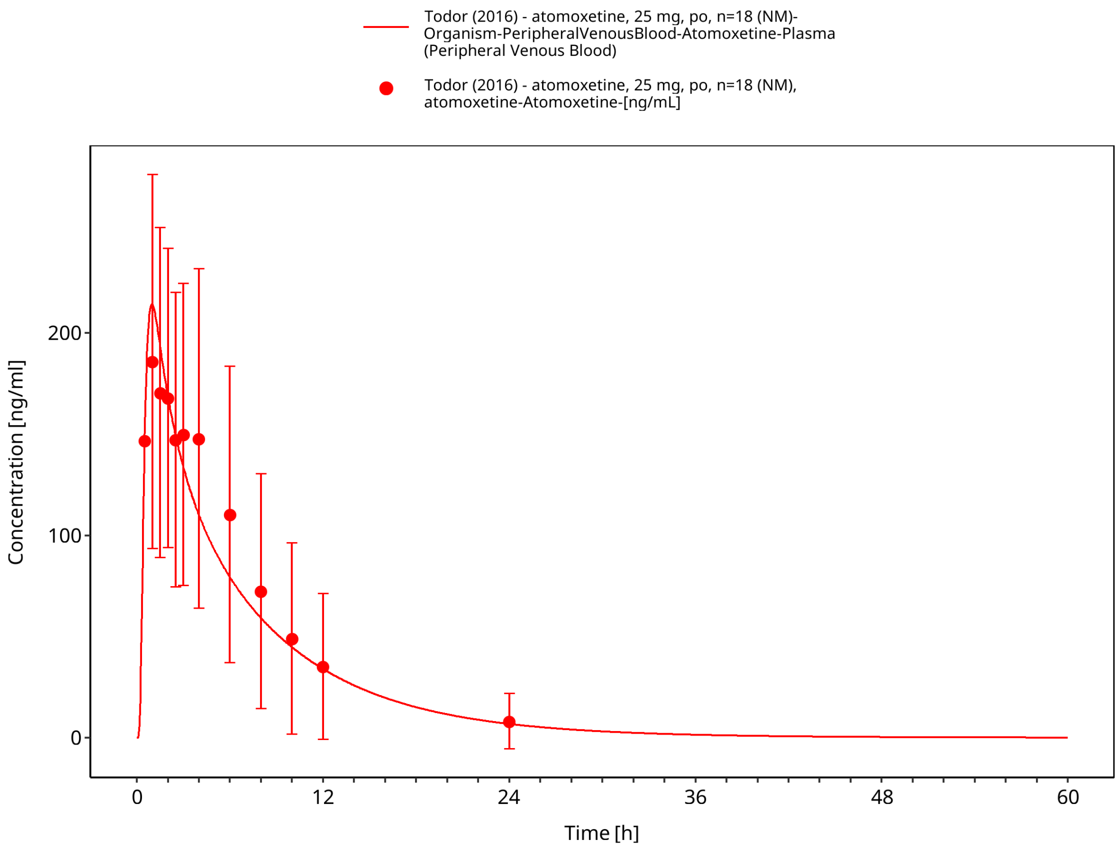

Figure 3-13: Time Profile Analysis

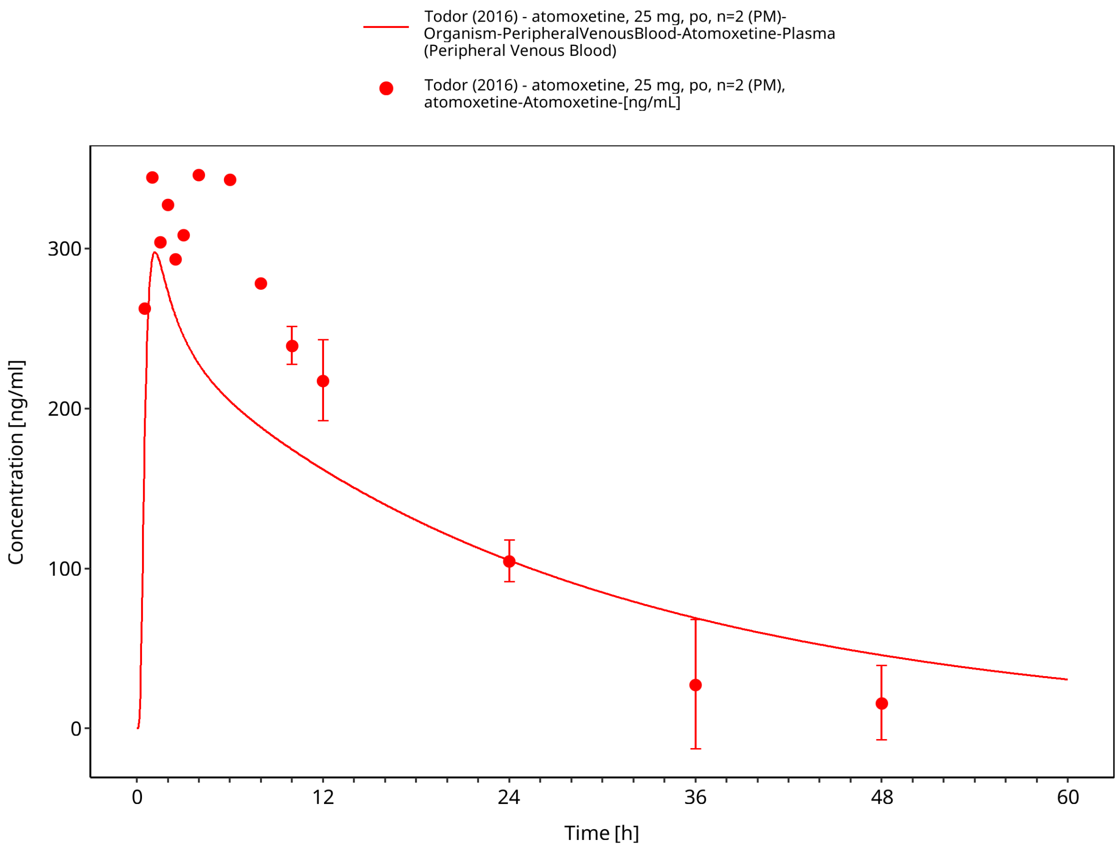

Figure 3-14: Time Profile Analysis

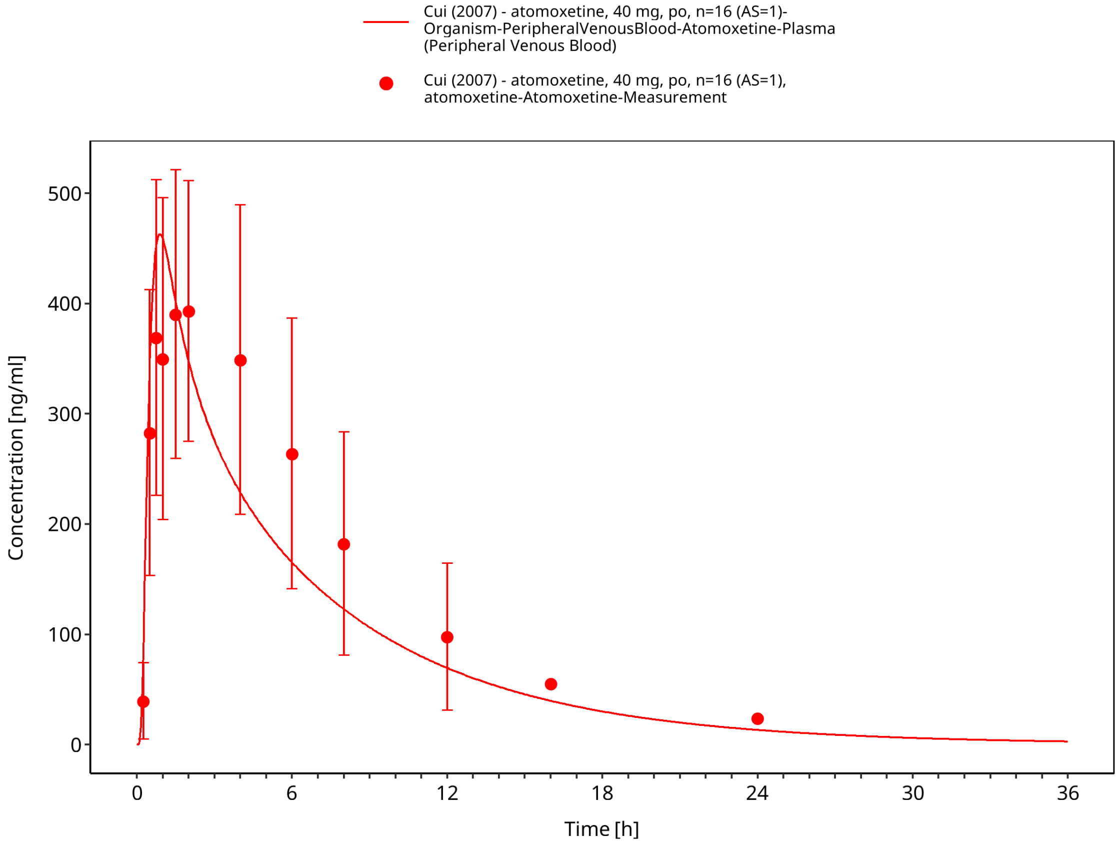

Figure 3-15: Time Profile Analysis

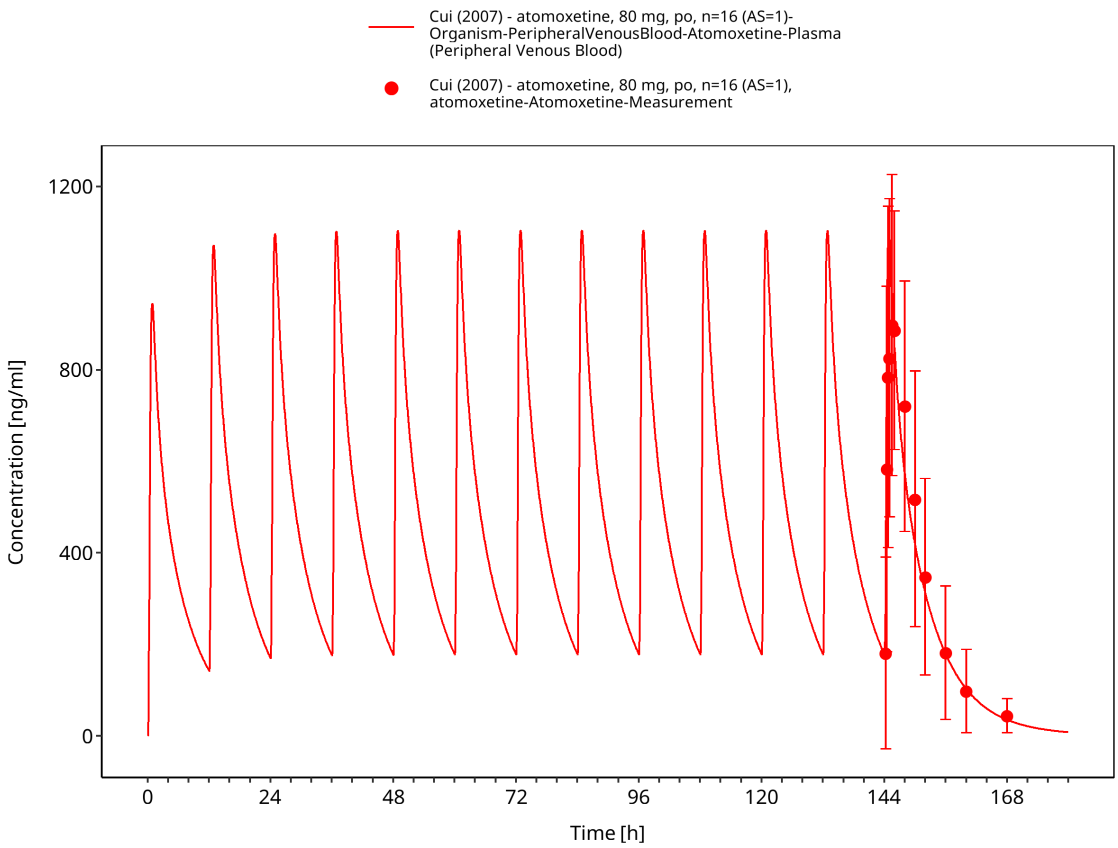

Figure 3-16: Time Profile Analysis

4 Conclusion

The presented PBPK model adequately describes the oral pharmacokinetics of atomoxetine in the evaluated adult CYP2D6 phenotype and activity-score groups.

5 References

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